Bonus Guide 4 of 5

THE POTTERY TEMPLATE VAULT

Glaze Layering Guide

Combining glazes layer by layer for depth, texture & unique effects

Glaze Layering Guide

> Layering two or more glazes on the same piece creates surface effects impossible to achieve with a single glaze. When glazes interact at high temperature, they produce unexpected textures, color breaks, movement and depth. This guide teaches you to do it systematically rather than by accident.

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1 · How Glaze Layering Works

At firing temperature, glazes melt and flow. When two glazes share the same surface, their molten layers mix at the interface, creating a third "zone" that is neither glaze alone. The result depends on:

- **Which glaze is on top vs. bottom** — order matters; the same pair in reverse often gives a different result.
- **The combined thickness** — two thin coats behave differently from two thick coats.
- **The fluxes in each glaze** — high-flux (fluid) glazes move more; low-flux (stiff) glazes anchor the combination.
- **The firing temperature** — higher temperatures increase movement and mixing.

2 · Safety and Thickness Rules

> ⚠️ **Warning**
>
> Layered glazes are thicker than single-glaze applications. If either glaze is fluid at firing temperature, the combined layer can run off the piece and permanently fuse to your kiln shelf. Always use a catch tray or kiln furniture under highly fluid combinations, and always fire a test tile first.

Thickness guidelines

- Total combined thickness: no more than **2.5–3 mm** on vertical surfaces.
- On horizontal surfaces (plates, serving boards): reduce by 20%.
- Always wax the foot ring before any layered application.
- Leave the bottom 10–15 mm of any vessel ungrazed or apply very thinly.

3 · Choosing Which Glazes to Layer

Complementary pairs (tend to work well)

- **Stiff base + fluid top** — the fluid glaze moves and pools in the lower areas while the stiff glaze anchors the combination. Creates beautiful movement.
- **Matte base + gloss top** — the matte smooths the surface while the gloss adds translucency. The boundary between them produces a semi-matte effect.
- **Dark base + light top** — where the top glaze is thin, the dark base shows through; where it pools, it appears lighter.

Combinations to avoid

- Two highly fluid glazes together — likely to run badly and cause damage.
- Zinc-containing glazes + chrome-containing glazes — they react to produce an unpleasant brown that is difficult to control.
- Glazes with very different thermal expansion coefficients — may cause severe crazing or shivering (glaze pops off).

4 · Layering Methods

Method A: Dip-over-dip

1. Dip the piece in glaze 1 (base). Let dry completely.
2. Dip in glaze 2 (top) — hold briefly and withdraw.
3. Clean the foot and fire.

Best for full-coverage layering on the whole piece.

Method B: Brush-over-dip

1. Dip the piece in glaze 1. Let dry.
2. Brush glaze 2 selectively — on the upper third, in horizontal bands, or in a drip pattern starting from the rim.

Best for creating a graduated or banded effect.

Method C: Pour layering

1. Pour glaze 1 over the outside while rotating. Let dry.

2. Pour glaze 2 from the rim in one side, rotate the piece so it flows partially around the exterior.

Creates organic, waterfall-like patterns. No two pieces are identical.

Method D: Finger-wipe resist

1. Apply glaze 1 all over. Let dry.
2. Wipe off glaze 1 in vertical stripes with a damp finger — exposing the bisque.
3. Apply glaze 2 over everything. It now covers the bisque stripes and layers over the remaining glaze 1.

Creates a striped pattern where alternating zones are single- and double-glazed.

5 · Five Classic Combinations to Start With

These are starting points — test on tiles before applying to finished pieces.

#	Base (1st)	Top (2nd)	Expected result
1	Tenmoku (dark, iron-rich)	Celadon (clear green)	Dark base with green pooling in low areas; rust-orange breaks on edges and texture
2	White matte	Cobalt blue gloss	Soft blue-white with depth; blue intensifies where thicker
3	Black matte	Chun blue (opalescent)	Dramatic blue over dark base; milky pooling in recesses
4	Iron red	Clear gloss	Clear amplifies the red; where thin, red saturates; where thick, slight yellowing
5	Shino (white-orange)	Ash grey (fluid)	Ash flows and breaks; Shino shows through as warm patches; heavy carbon-trap texture

> ⭐ **Important**
> The order "base then top" always means base is applied first (touching the clay) and top is applied over it. Reversing the order with the same two glazes often gives a completely different result — worth testing both directions.

6 · Reading a Fired Layered Tile

After firing a test tile, look for these zones:

- **Single-glaze zone** — where only one glaze is present. Compare this to your standard single-glaze test.
- **Overlap zone (center)** — the primary interaction area. Note color, texture, movement.
- **Edge zone** — where one glaze thins out. Often the most beautiful, with color breaks and thinning effects.

- **Run zone (bottom)** — shows how fluid the combination is. The extent of running tells you whether you need to reduce thickness or leave a wider unglazed margin.

7 · Building Your Own Combinations Systematically

The 3×3 test matrix

Choose 3 glazes (A, B, C). Create tiles for all 6 combinations (A over B, B over A, A over C, C over A, B over C, C over B) plus 3 single-glaze tiles. Fire together. This gives you 9 data points and you can map exactly what each pair does in each direction.

Recording your results

Keep a glaze journal. For each combination, note:

- Glaze names and brand
- Application method and estimated thickness
- Firing cone and kiln atmosphere
- Photo of the fired tile
- Description: color, texture, movement, food safety concerns
- Decision: use as is / adjust thickness / test reversed order / discard

8 · Common Problems and Fixes

Problem	Cause	Fix
Combination runs badly	Both glazes too fluid; combined layer too thick	Reduce thickness; try stiff base under fluid top; use taller catch tray
No interaction — looks like two separate glazes	Both glazes are very stiff; temperature too low	Try a more fluid top glaze; increase temperature slightly (test carefully)
Colors look muddy together	Colorant clash (e.g., cobalt + iron in equal strength)	Try a neutral base (white, clear) under one color, or reduce one colorant's percentage
Top glaze crawls off the bottom glaze	Top glaze shrinks more than the base; base too powdery/dusty	Apply base more thinly; fire base glaze to bisque first before adding top
Combination looks great on the tile, terrible on the pot	Gravity changes on a 3D form; vertical vs. horizontal glaze behavior differs	Test on a vertical test cylinder as well as a flat tile before committing

Layering checklist

- Did I test this combination on a tile first?
- Is the total combined thickness within the safe range (≤3 mm on verticals)?
- Is the foot ring waxed and clean?
- Have I left enough unglazed margin at the base?
- Did I use a catch tray under the piece in the kiln?

Next: Firing Guide — bisque and glaze firing temperatures, kiln loading, schedules and common firing errors.